

Digital Logic Design Report

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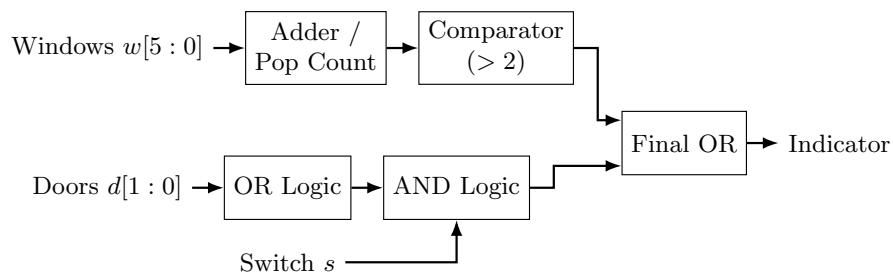
1 Problem 1: Automation Alarm System

1.1 Problem Description

An automation system utilizes sensors to monitor six windows, two doors, and a safety switch. A combinational circuit is required to trigger an indicator light if either of the following conditions is met: 1. Any door is open while the safety switch is activated. 2. More than two windows are open simultaneously.

1.2 Solution Architecture

The solution is implemented using two architectures: Dataflow and Behavioral. In both cases, the core logic requires summing the high bits from the six window sensors and comparing the result to 2. Simultaneously, the two door sensors are passed through a logical OR gate, and the result is ANDed with the safety switch. The outputs of these two main logical branches are evaluated through a final OR gate to drive the indicator.



2 Problem 2: Sensor Threshold System

2.1 Problem Description

A system reads four 4-bit sensors. A 2-bit address selector dictates which sensor reading is active. If the selected reading is greater than or equal to a user-defined

4-bit threshold, an LED indicator turns on.

2.2 Solution Architecture

The design pairs a Multiplexer (MUX) with a Digital Comparator. The 2-bit address selector (**sel**) drives a 4-to-1 MUX to isolate one of the four 4-bit sensor inputs (**s0**, **s1**, **s2**, **s3**). The 4-bit output of the MUX is immediately routed into an arithmetic comparator. The comparator evaluates if the MUX output is $\geq$ the 4-bit **thresh** input, driving the LED high if true.

